

Aim: Implement Binary Search to find an item in an ordered list.

1. Imagine you are developing a treasure hunt game where player must find a specific hidden item in a series of ordered item in the box each item is label with the string and arrange in alphabetical order. WAP program in C++ that imployes an efficient (binary)search to help the players search for item in the box.

CODE:

```
#include <iostream>

#include <cstring>

using namespace std;

int randomTreasure(int n,string item,string t[]){

    int l=0;

    int h=n-1;

    while(h>=l){

        int m=(l+h)/2;

        if(t[m]==item){

            cout<<"Congratulations you got the correct item:"<<item<<" from the

treasure box!!!";

            return 0;

        }

        if(t[m]>item){

            h=m-1;

        }

        if(t[m]<item){

            l=m+1;

        }

    }

    cout<<"Sorry!, Better luck next time :)";

    return 1;

}

int main(){

    int n;

    cout<<"Enter the number you want to add in the treasure box:";
```

```
    cin>>n;

    string t[n];

    cout<<"Enter the items in the box:";

    for(int i=0;i<n;i++){

        cin>>t[i];

    }

    string item;

    cout<<"Enter the name of any one item:";

    cin>>item;

    randomTreasure(n,item,t);

    return 0;

}
```

OUTPUT:

```
Enter the number of items you want to add in the treasure box:5
Enter the items in the box:Diamond Emerald Netherite Ruby UC
Enter the name of any one item:UC
Congratulations you got the correct item:UC from the treasure box!!
-----
Process exited after 54.75 seconds with return value 0
Press any key to continue . . . |
```

```
Enter the number of items you want to add in the treasure box:5
Enter the items in the box:Diamond Emerald Netherite Ruby UC
Enter the name of any one item:Gold
Sorry!, Better luck next time :)
-----
Process exited after 27.56 seconds with return value 0
Press any key to continue . . . |
```